

Can current surveillance follow doxy-PEP-associated *Staphylococcus aureus* resistance? A multi-stream evidence audit

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Running title: Surveillance and doxy-PEP bystander *S. aureus* resistance

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Synopsis

Background. Doxycycline post-exposure prophylaxis (doxy-PEP) prevents bacterial sexually transmitted infections and has scaled into guidance internationally. Doxycycline also selects tetracycline resistance beyond its target, and *Staphylococcus aureus*—a commensal and pathogen for which doxycycline is one of a limited number of oral options—is the natural bystander.

Objectives. To determine whether the guidance and surveillance deployed around doxy-PEP could follow the bystander-resistance signal the pivotal trial's final analysis resolved, rather than to estimate a population effect.

Methods. A reproducible, public-data audit of three evidence streams—clinical trials, governmental guidance and deployed US surveillance—each scored against one yardstick: the within-exposed relative risk it could detect, benchmarked to an empirical colonisation estimate.

Results. The final randomised DoxyPEP analysis reported a 3.89-fold higher hazard of incident doxycycline-resistant *S. aureus* under doxy-PEP (95% CI 1.42–10.68; 68/393 versus 5/163), with no effect on colonisation clearance. Yet 0 of 6 outbreak-matched governmental guidelines

(0 of 10 coded) required *S. aureus* monitoring, and 0 of 12 established US surveillance systems linked doxy-PEP exposure to an *S. aureus* tetracycline phenotype at a common population denominator. Under realistic exposure density and isolate volumes, an ordinary population system would need an implausible within-exposed effect ($RR \approx 14\text{--}64$) to move a rate.

Conclusions. The upstream trial generated information the downstream architecture is not built to receive. Following the signal requires sentinel surveillance linking doxy-PEP exposure to the *S. aureus* tetracycline phenotype at a common denominator. No population-level causal claim is made.

Introduction

Doxycycline post-exposure prophylaxis (doxy-PEP) substantially reduces bacterial sexually transmitted infections (STIs) and has moved quickly from trial result into guidance at every level, from the World Health Organization through national and municipal health departments.^{1,2} Every dose is selective pressure, and not only on the organisms it targets. *Staphylococcus aureus* is the consequential bystander: it colonises the anatomical sites these trials swabbed and carries mobile tetracycline resistance determinants.³ Doxycycline is one of a limited number of oral options for outpatient skin and soft-tissue infection, a choice often made empirically before any methicillin result returns,^{4,5} so tetracycline susceptibility can matter clinically before methicillin status is known.

Clinicians who prescribe doxy-PEP have registered the concern directly: in a survey of Italian STI-clinic dermatologists, 91.7% named methicillin-resistant *S. aureus* (MRSA) selection as a worry, tied with gonococcal resistance as the highest-rated.⁶ The trial literature has now answered a version of it. In the final DoxyPEP analysis, among participants free of doxycycline-resistant *S. aureus* at baseline, doxy-PEP carried a 3.89-fold higher hazard of incident doxycycline-resistant *S. aureus* (95% CI 1.42–10.68; 68/393 versus 5/163; $P = 0.0044$), while the hazard of clearing colonisation did not differ between arms (HR 1.01, 0.69–1.46).⁷ The trial's authors called for public-health surveillance of antimicrobial resistance (AMR) in *S. aureus* and other commensals.

That call defines the present question. We ask not whether doxy-PEP drives population-level resistance—a claim we do not make—but whether the guidance and surveillance deployed around doxy-PEP could follow the signal the trial resolved, at the population level where the clinical stake lies. The clinically decisive axis is tetracycline (doxycycline) susceptibility used in empirical oral therapy, not the methicillin axis around which staphylococcal surveillance is organised. We audit the three places an answer would have to come from—the trials, the guidance and the deployed surveillance—each against a single detectability yardstick.

Materials and methods

Design

The study is a documentary and quasi-computational audit assembled in three streams, collected and analysed in sequence: clinical trials (Stream A), the governmental guidance built on them (Stream B) and the population surveillance meant to monitor them (Stream C). No individual-level or restricted data were used; every input is public, so no ethical approval was required. Sources were located, read, coded with page/section/table locators and SHA-256-pinned before analysis; analysis decisions that could go either way were date-logged before results were known; and every figure and table regenerates from a single command over a tested pipeline archived with a citable DOI.⁸ A large language model assisted with writing and analysis engineering; the author verified every numerical value and quotation against primary sources and takes full responsibility (see Acknowledgements).

Detectability yardstick

Each stream is scored on one commensurable quantity: the within-exposed relative risk (RR) it could detect. The empirical benchmark is Soge *et al.*⁹ (tetracycline-resistant *S. aureus* colonisation, 18% versus 8% in doxy-PEP users with >3 doses/month versus non-users; $RR \approx 2.25$); a cross-organism gonococcal $RR \approx 1.42$ is carried as a flagged, more conservative secondary.

Stream A—trials

For each primary-trial *S. aureus* arm comparison (first-party published data only), we computed, from arm sizes and observed control rate, the exact (Fisher) power to detect the externally fixed benchmark effect and the minimum detectable RR at 80% power; fixing the effect externally makes this a design-based sensitivity analysis rather than observed power. The *S. aureus* assay is doxycycline throughout (E-test, MIC $\geq$ 16 mg/L). We tabulated how the resistance result was reported across trial venues and secondary syntheses (numerator, denominator basis, conditioning set and terminology). The final trial performed the appropriate participant-level time-to-event test itself; we cite that randomised result directly and did not re-estimate it. Cross-sectional carriage cannot resolve the clustered transmission structure of *S. aureus* in these networks; the supporting dispersion, trend-test degradation and outbreak-shape analyses are MRSA-scoped (MRSA being a subset, not the analytic unit) and are reported in the Supplementary data (available as Supplementary data at JAC Online).

Stream B—guidance

To make the denominator principled, we anchored it to the same clustering evidence used in Stream A: US jurisdictions with documented community-associated MRSA/USA300 transmission in men who have sex with men.^{10,11} We took governmental public-health guidance only (clinic and provider-organisation documents excluded, with reasons). Each document was coded from full text, one record per document, every finding carrying a locator, for whether it requires, suggests or omits *S. aureus* monitoring and whether it names, counsels on or measures the organism. A blinded second coder double-coded, with per-field Cohen's κ . An independence check separated genuinely independent declines from documents that defer to, or are adapted from, one another.

Stream C—surveillance

We enumerated a closed universe of established US governmental surveillance systems (federal, state or metropolitan) that could plausibly capture, at a population denominator, either

doxy-PEP/doxycycline exposure or *S. aureus* AMR phenotype, and coded each on three capabilities: exposure-side capture, phenotype-side capture (of the *S. aureus* tetracycline unit, not the MRSA subset) and the join, recording the geographic resolution of each side. A supporting analysis asked whether, even at a common denominator, the exposed subgroup is dense enough to move a population rate. The dilution fraction is $f = \phi(\text{male-PrEP rate}/10^5) \text{ uptake } \kappa$, population-independent because it tracks exposure density, not headcount, and the minimum detectable change in a proportion near baseline R_0 inverts to $RR_{\text{needed}} = 1 + \text{MDE}/(fR_0)$. Parameters were bounded to favour detection (so a failed gate is robust); full grids and the metropolitan break-even are in the Supplementary data.

Results

Stream A: the trials resolved the signal but reported it unstably

When the analysis matched the question, the signal appeared. The final DoxyPEP analysis ran the participant-level test the interim had not—time to first detection of doxycycline-resistant *S. aureus* by arm, standard-of-care censored at crossover, by Cox proportional hazards—and returned HR 3.89 (95% CI 1.42–10.68; $P = 0.0044$), with colonisation clearance unchanged (HR 1.01).⁷ The 2023 interim, by contrast, read a cross-sectional resistance fraction as reassurance and was underpowered for any externally specified effect (0 of 4 comparisons powered; minimum detectable RR 3.8–5.1 against the benchmark 1.42–2.25). The hazard ratio is not comparable to that cross-sectional minimum: the estimand, not the sample size alone, became adequate to the question.

Two measurement features of the interim record persist and matter downstream. First, a unit mismatch: the trial’s data table is headed “phenotypic resistance to the tetracycline antibiotic class” while its methods define the *S. aureus* assay as doxycycline, and the federal guideline then describes the doxycycline result as “tetracycline resistance in *S. aureus*”.² Because efflux and ribosomal-protection mechanisms differ in whether doxycycline still works (below), an endpoint scored as “the tetracycline class” does not answer the treatment-relevant question. Second, the signal was reported over differing, incompletely documented denominators and

then propagated into secondary reporting under further contrasts (Table 1): the resistant numerator changes from 5 to 16 to 28 and the denominator basis shifts from colonised carriers to all-swabbed, while the percentage travels more reliably than the denominator, contrast and terminology needed to interpret it. The individual-level data that could reconcile these first-party reports are not released.¹ Even the intensively surveilled gonococcus was phenotyped in fewer than one diagnosis in five (resistance testing unavailable for 83%, 212/256).¹²

Across trials the phenotype was measured on incompatible terms: DoxyPEP measured doxycycline resistance by E-test over two denominators; DuDHS measured it by disc diffusion in single-digit carriers; DOXYVAC measured only MRSA carriage—a methicillin, not a tetracycline, quantity, and one that itself rose within the doxy-PEP arm (1.8% to 9.9%) while being reported as a between-arms null.¹³ A pooled estimate is not defensible, and the heterogeneity is itself a finding. The trial resolved the doxycycline phenotype but not its molecular basis—efflux (*tet(K)*) versus ribosomal protection (*tet(M)*)—which a single breakpoint cannot separate.^{3,5}

Stream B: guidance names the risk but instruments nothing

The governmental guidance shows a consistent within-document asymmetry: the efficacy question received a graded, systematic evidence review, while the AMR-harm question—explicitly including resistant-pathogen development—received an ungraded review in the same document. Of the six documented-outbreak jurisdictions with governmental doxy-PEP guidance (San Francisco, Chicago, New York City, Boston/Massachusetts, Los Angeles County and San Diego), 0 require or suggest *S. aureus* monitoring. The decline is not an artefact of copying: four jurisdictions are independently authored, one (Los Angeles County) is adapted from San Francisco and one (Massachusetts) from an external template. Nor is it for want of naming the organism—three of the six name *S. aureus* specifically, two invoke only a generic “microbiome” risk and one is silent; even where named, none measure it. Across the full coded corpus (10 documents: CDC, WHO and eight US jurisdictions) the result is identical: 0 instrument the bystander. Inter-coder agreement was 92%; the single field with meaningful disagreement ($\kappa = 0.58$) is reported as such. Providers are nonetheless required to counsel patients about resistance in commensal organisms—a harm no system measures.

Stream C: no deployed system links exposure to the phenotype

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Following the signal requires one system to carry, at a common population denominator, both
who is exposed to doxy-PEP and what tetracycline phenotype *S. aureus* carries. Across the
closed universe of twelve established US systems, none does (Figure 1). Two systems capture
exposure at a population denominator (AIDSVu and CDC AtlasPlus, as a pre-exposure pro-
phylaxis proxy, not doxy-PEP itself),¹⁶ and zero capture *S. aureus* tetracycline resistance at a
population denominator: the join is vacuous because one half does not exist.

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The absence is a codified seam. Wherever *S. aureus* is surveilled at a population denomina-
tor the reported axis is methicillin: the one population-based platform, Active Bacterial Core
surveillance/Emerging Infections Program (ABCs/EIP), reports invasive MRSA; the National
Healthcare Safety Network reports MRSA events; and the only routine source of *S. aureus*
tetracycline susceptibility, the facility cumulative antibiogram, is non-standardised and carries
no population denominator, because CLSI M39 mandates the MRSA/MSSA split and makes
tetracycline a selectively suppressible supplemental agent. The National Antimicrobial Re-
sistance Monitoring System excludes *S. aureus* altogether; tetracycline-resistant *S. aureus* is
nationally notifiable nowhere. Exposure exists at county/metro resolution and phenotype only
at denominator-free facility level, so no geography carries both. A system could be built—most
directly by adding *S. aureus* tetracycline susceptibility to ABCs/EIP, which already holds the
denominator—but none is deployed.

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Nor could an ordinary population system recover the missing side by collecting more iso-
lates. Under realistic exposure density and isolate volumes the within-exposed effect required
to move a population rate runs to $RR \approx 14\text{--}64$ across the design-effect range—far above any
plausible effect (Figure 2)—and detection appears only in a best-case cell that compounds
four simultaneously implausible assumptions ($RR \approx 1.93$, and failing on any one assumption
alone). The metropolitan break-even requires a male-PrEP density 2.1–33 times the densest US
geography, or roughly 325% of adult males on PrEP under realistic isolate volumes. The only
configuration that restores sensitivity is to sample *S. aureus* directly from doxy-PEP-exposed
persons, which converts surveillance into a sentinel cohort.

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Alt text: A matrix of twelve US surveillance systems against exposure and phenotype capabil-

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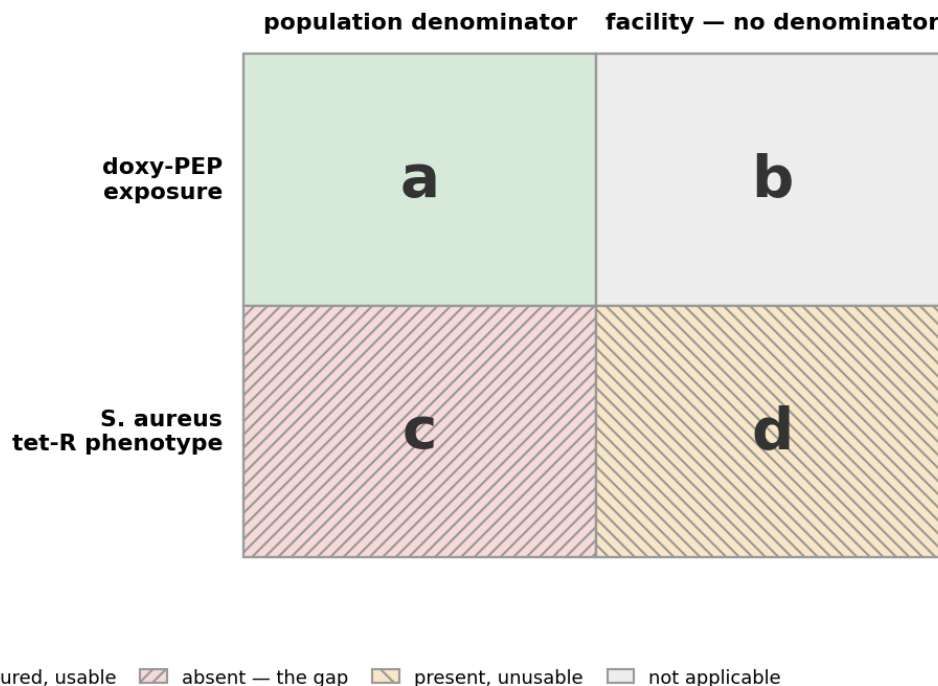


Figure 1: No deployed US system links doxy-PEP exposure to *S. aureus* tetracycline resistance at a common population denominator. Linking the two requires one column usable in both the exposure and phenotype rows; none is. Exposure is captured only as a PrEP proxy (AIDSVu, CDC AtlasPlus); the *S. aureus* tetracycline phenotype is captured at a population denominator nowhere, because the one population-based platform reports the methicillin (MRSA) subset. Of 12 systems, 0 join the two.

ity; shading shows that exposure is captured only as a PrEP proxy and the *S. aureus* tetracycline phenotype is captured at a population denominator nowhere, so no column links both rows.

Alt text: A logarithmic relative-risk axis with a shaded benchmark band at 1.42 to 2.25; the trial hazard ratio 3.89 sits above the band, surveillance detectability thresholds of about 14 to 64 sit far to the right, and guidance has no threshold.

The literature mirrors the instrumentation

A defined PubMed search (2015–2026) identified 361 doxy-PEP papers, of which 13 (3.6%) mention a bystander staphylococcal organism and only 2 measure tetracycline-resistant *S. aureus* against a doxy-PEP exposure contrast. The bystander is rarely mentioned and, when mentioned, rarely measured against exposure—the research attention concentrated where the existing instruments already point. Table 2 summarises the three streams against the yardstick.

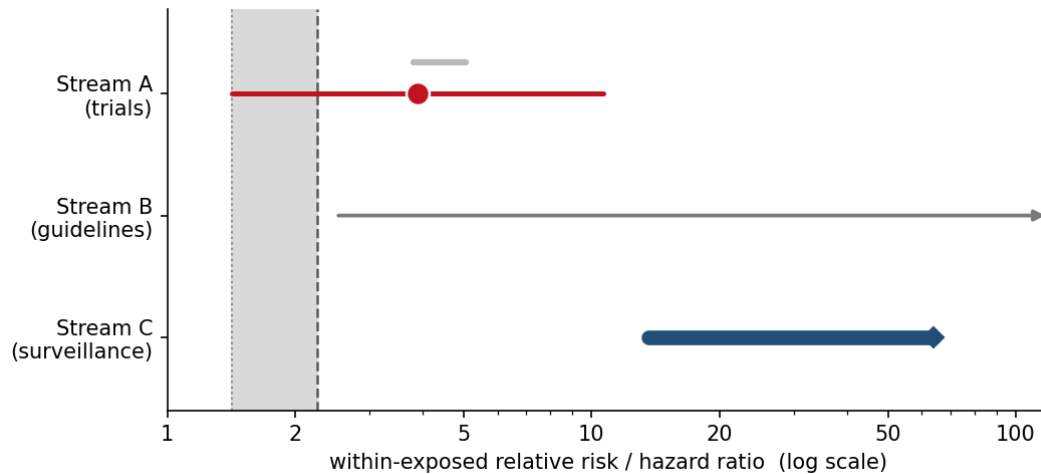


Figure 2: One axis—within-exposed relative risk (log scale)—carries all three streams against the plausible benchmark band (RR 1.42–2.25). The trial’s resolved effect (HR 3.89) clears the band; guidance requires no *S. aureus* measurement so no detection threshold exists; and deployed surveillance would need RR ≈ 14–64 to move a population rate. The signal is above the line while the systems built to follow it are not.

Discussion

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The pivotal trial has now resolved the bystander question it was long thought unable to reach: a significant increase in incident doxycycline-resistant *S. aureus* under randomised exposure, with no matching effect on carriage, and a call from its authors for surveillance of exactly this.⁷ The upstream instrument generated the information. Across the two systems that would have to receive it—guidance and deployed surveillance—the answer is that they cannot: guidance instruments nothing (0/6; 0/10) and no deployed system joins exposure to the *S. aureus* tetracycline phenotype at a common denominator (0/12).

The sharper reading is that the trial’s own measurement system evolved to meet its internal question while the implementation system did not. The 2023 interim read a cross-sectional fraction as reassurance; the 2024 guideline appended a call for monitoring; the 2025 final analysis asked the estimand the others had not and returned HR 3.89. Three years, one underlying phenotype, an evolving estimand¹⁷—and, at the end, a demonstrated signal handed to systems built to measure something else.

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The phenotype axis is the wrong one for the clinical question

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Staphylococcal surveillance is standardised around methicillin at every tier, yet the treatment- 180
relevant axis here is tetracycline susceptibility, used empirically before methicillin status is 181
known. The two are not interchangeable, and neither is the molecular basis of tetracycline 182
resistance: *tet(K)* efflux leaves doxycycline largely usable, whereas *tet(M)* ribosomal protection 183
can compromise it, and the trial resolved incident doxycycline resistance without resolving 184
which.^{3,5} A surveillance variable can remain valid for the question it was designed to answer 185
while being insufficient for the one now imposed on it. We do not forecast which mechanism 186
is spreading; the point is that the inherited variable cannot say. 187

What instrument would answer the question

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Following the signal to the population requires an instrument the deployed architecture does 189
not provide: sentinel sampling of *S. aureus* from doxy-PEP-exposed individuals, which raises 190
the exposed fraction by construction and fixes the denominator; resistance reported per carrier 191
and by mechanism (*tet(M)* versus *tet(K)*) rather than as a pooled tetracycline-class rate; dosing 192
frequency rather than prescription status; a feed carrying *S. aureus* tetracycline susceptibility at 193
a population denominator, most directly by adding it to the platform that already tracks invasive 194
MRSA and joining it to exposure on geography; and release of the individual-level trial data. 195
None is exotic; none is in place at scale. 196

Measurement inheritance

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These failures share a structure. A guideline or a surveillance system carries in its sampling 198
frame, cadence, coding, denominator and unit the organism and question it was built for. The 199
doxy-PEP apparatus was built to measure gonococcal efficacy; applied to a bystander com- 200
mensal it inherits a denominator the intervention itself moves, a phenotype label spanning two 201
mechanisms, and a surveillance architecture standardised around methicillin while the tetracy- 202
cline unit falls orthogonally through it. We call this *measurement inheritance*: a downstream 203
system retains the dimensions—estimand, denominator, unit, cadence, linkage, measurement 204

responsibility—appropriate to the upstream question but not those the downstream question re- 205
quires, so the question becomes unanswerable not for want of data but for want of an inherited 206
dimension. It is built from familiar parts—construct validity, estimand mismatch, ascertain- 207
ment and surveillance bias,¹⁸ metric substitution of the Goodhart type¹⁹—but concerns how 208
such mismatches couple *across* stages of an evidence chain. The downstream gap is there- 209
fore not closable by a better analysis of the trial’s data—that analysis has been done, and it is 210
significant—but is a property of the guidance and surveillance instruments. 211

The coupling has a distributive consequence, and it is a measurement asymmetry before it is 212
a moral one. The benefit is measured in an identified exposed population, while any resistance 213
externality would fall on the outpatient with a skin infection and, at the sharp end, the dialysis, 214
surgical or oncology patient for whom an oral anti-staphylococcal agent is no convenience— 215
none party to the prescribing decision or counted by any deployed system.²⁰ The evidence 216
architecture permits one side of the ledger to be estimated far more completely than the other; 217
a negative externality no instrument can price is not thereby small, only unweighable. 218

Limitations 219

We make no population-level causal claim and no transmission claim: whether the individual- 220
level signal propagates into a population externality is precisely what no deployed system can 221
determine. The trial’s own randomised result is significant and we accept rather than dispute 222
it; our earlier aggregate reconstruction of the interim is retired in its favour. Colonisation 223
is not infection, and the colonisation–infection link is itself unmeasured here. Much of the 224
evidence for how *S. aureus* clusters and moves through these networks exists only for its MRSA 225
subset, so where our clustering analysis relies on MRSA we mark it as such—an instance 226
of the inherited limitation, not an exception to it. Finally, the trial reaches only a roughly 227
twelve-month horizon; persistence, dose–response and mechanism remain open. Consistent 228
with a falsification standard set in advance, the inheritance claim fails if a doxy-PEP guideline 229
comes to require *S. aureus* monitoring, or if a deployed system links doxy-PEP exposure to an 230
S. aureus tetracycline phenotype at a common population denominator. 231

Conclusions

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The randomised instrument generated a bystander-resistance signal that the guidance and surveillance built around doxy-PEP are not instrumented to follow. The gap is one of measurement architecture, not of missing data, and it is closable—by sentinel surveillance that links doxy-PEP exposure to the *S. aureus* tetracycline phenotype at a common denominator. Until it is built, the population question the trial’s authors posed cannot be run.

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Transparency declarations

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Table 1: Propagation of the DoxyPEP *S. aureus* resistance signal across reporting venues. Rows are not assumed to be identical estimands; differences in contrast, conditioning set, denominator and terminology are the object of comparison.

Venue	Reported result	Frame / conditioning set
NEJM appendix ¹	5/31 = 16%	Month-12 doxy-PEP arm; resistant among carriers
CROI abstract ¹²	16/137 = 12%	Doxy-PEP arm; all-swabbed denominator
CDC guideline ²	28/222 = 13%; summarised 5%→13%	Doxy-PEP arm; guideline frame
JAMA Insights ¹⁴	5% versus 4%	Doxy-PEP versus control; “infected with”
JAMA Synopsis ¹⁵	5%→16%	Within arm, among <i>S. aureus</i> -isolated

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Table 2: Headline audit of the doxy-PEP evidence base against one detectability yardstick. Each cell is licensed by the Results above; no composite score is implied.

Stream	Question	Result
A. Trials	Was the bystander signal resolved?	Yes—HR 3.89 (1.42–10.68), incident doxy-R <i>S. aureus</i>
B. Guidance	Is <i>S. aureus</i> monitoring required?	No—0/6 outbreak-matched; 0/10 coded
C. Surveillance	Is exposure linked to phenotype?	No—0/12 systems; $RR_{\text{needed}} \approx 14\text{--}64$ to detect